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Practice

Practice session 1

Q. Sentence 1 : "The cat sat on the mat". Find the cosine similarity.
2 : "The cat is sitting on the mat"

→ tf-vector : { the cat sat on mat is sitting }

S1 :	2	1	1	1	1	0	0
2 :	2	1	0	1	1	1	1

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{\|a\|\,\|b\|} = \frac{7}{\sqrt{8}\,\sqrt{9}} = \frac{7}{3\sqrt{8}} = 0.8249$$

Q. Sentence 1 : "climate change is affecting global weather patterns". Find the Jaccard similarity.
2 : "global weather patterns are being affected by climate change".

→ $\frac{|A \cap B|}{|A \cup B|} = \frac{|\{\text{climate, change, global, wear, pat}\}|}{|\{\text{clim, ch, is, aff, gl, w, p, are, being, affd, by}\}|} = \frac{5}{11} = 0.4545$

Q. Sentence 1: Artificial intelligence is transforming various industries. Find AM & cosine similarity and Jaccard similarity.
sentence 2: Various ind. are being transformed by art. int.

→

Jacc_sim = $\frac{4}{10} = 0.40$

cosine-sim :

Art	int	is	trans.	var	ind.	are	being	trans	by
1	1	1	1	1	1	0	0	0	0
1	1	0	0	1	1	1	1	1	1

$$\text{cos-sim} = \frac{4}{\sqrt{6}\,\sqrt{8}} = \frac{4}{\sqrt{48}} = \frac{1}{\sqrt{3}} = 0.5773$$

$$\text{AM} = \frac{0.4 + 0.5773}{2} = 0.48865 \qquad \frac{0.4 + \frac{4}{\sqrt{48}}}{2} = 0.48867$$

Q. use bi-gram, not unigram.
S1: benefits of healthy eating are widely recognized.
S2: healthy eating habits offer numerous adv.

Find weighted avg of cosine-sim for 1gram and 2 gram. (wts : 0.4, 0.6 resp)

→ Bi-gram

tf vector : [benefits A] [of healthy] [healthy eating] [eating are] [are widely] [widely recognized] [eating habits] [habits offer] [of num] [num adv]

1	1	1	1	1	1	0	0	0	0	0
0	0	1	0	0	0	1	1	1	1	1

$$\text{cos-sim} = \frac{1}{\sqrt{6}\,\sqrt{5}} = \frac{1}{\sqrt{30}}$$

Uni-gram :

Benef.	of	heal	eat	are	wide	recog.	habits	offer	num	adv
1	1	1	1	1	1	1	0	0	0	0
0	0	1	1	0	0	0	1	1	1	1

$$\text{cos-sim} = \frac{2}{\sqrt{7}\,\sqrt{6}} = \frac{2}{\sqrt{42}}$$

$$\text{weighted avg} = (0.4)\left(\frac{2}{\sqrt{42}}\right) + (0.6)\left(\frac{1}{\sqrt{30}}\right) = 0.23298$$

Q.

Product ID	Sales (in $)	Cost (in $)	Profit (in $)
101	12.00	8.00	4.00
102	9.00	6.00	3.00
103	15.00	11.00	4.00
104	6.00	5.00	1.00
105	18.00	14.00	4.00

a) Find the maximum normalized z-score of the sample data for cost.
b) Find the mean normalized max-min value for sales.
c) In decimal scaling normalizn, what would the scaling factor be for all 3 attributes?

→ a) mean $= \frac{8+6+11+5+14}{5} = 8.8$ $\quad s = 3.701$ ← sample std. deviation

max z-score $= \frac{14-8.8}{3.701} = 1.405$

b) min-max for sales: $x' = \frac{x - \min}{\max - \min} = \frac{x-6}{18-6} = \frac{x-6}{12}$

$\frac{12-6}{12}, \frac{9-6}{12}, \frac{15-6}{12}, \frac{6-6}{12}, \frac{18-6}{12}$

$\frac{6}{12}, \frac{3}{12}, \frac{9}{12}, 0, 1$ $\quad$ mean $= \frac{1}{5}\left(\frac{1}{2}+\frac{1}{4}+\frac{3}{4}+1\right) = 0.5$

c) 2, 2, 1

Q. Consider a dataset:
22, 25, 55, 60, 29, 31, 45, 50, 42, 35
Perform equal-width binning (3 bins).
Calculate |bin2| + |bin 3| or ϕ(bin2) + ϕ(bin3)

→ 22, 25, 29, 31, 35, 42, 45, 50, 55, 60

Bin width $= \frac{60-22}{3} = \frac{38}{3} = 12.66$

[22 – 34.6) [34.6, 47.2) [47.2, 59.8)

22, 25, 29, 31 | 35, 42, 45 | 50, 55, 60

ϕ(bin2) + ϕ(bin3) = 6

Q. [55, 60, 25, 98, 68, 72, 90, 92, 75, 80, 88, 85]. Perform equal frequency binning with 4 bins. Find mean (bin3).

→ 25, 55, 60, 68, 72, 75, 80, 85, 88, 90, 92, 98

mean = 84.3333

of bins = 4
of items in each bin $= \frac{12}{4} = 3$